Exponential growth and decay

1

a

i Exponential growth

ii 0.6

b

i Exponential decay

ii 0.8

2 233

3 $f(t) = 26030 \times (1.011)^t$

4 $P(t) = 10\,637\,(1.0040)^t$

5

a $y = 750\,(1.03)^t$

b 869

6

a $y = 2900\,(1.09)^v$

b $\$5778.43$

7

a $y = 322\,000\,(0.95)^m$

b $\$249\,157.46$

8

a $y = 2310\,(0.974)^n$

b 1328

9

a $y = 200\,(3)^n$

b $\$48\,600$

10

a $y = 21\,(0.5)^n$

b 0.66 m

11

a $A = 300\,(0.5)^{\frac{t}{20}}$

b 0.59 grams

12

a Yes **b** No **c** No **d** Yes

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13 $r = 3.53\%$

14 631 bacteria

15

a 0.01

b $A = 38\,200 \times 0.99^n$

c \$35 964.54

d Motorbike

16 41 199

17 6591

18 262

19

a

Quarter(N)	1	2	3	4	5	6	7
Online Customers (C)	480	2880	17280	103 680	622 080	3 732 480	22 394 880

b $C = 480 \times 6^{N-1}$

c 29 023 764 480 sales

d To not assume that growth will change in exactly the same way in the coming quarters and try a different model.

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